

Algebraic Limits of the $\text{Cl}(6)$ Programme: A Schur Obstruction, the Forced Extension to $\text{Cl}(10)$, and Two No-Go Results for the Electroweak Sector

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Abstract

We delimit what the Clifford-algebra $\text{Cl}(6)$ approach to Standard-Model structure can and cannot determine, establishing three results. First, a *Schur obstruction*: any weak isospin $\text{SU}(2)$ commuting with color on the eight-dimensional $\text{Cl}(6)$ ideal acts only on the color singlets, leaving the quark triplet invariant. Quarks therefore cannot be weak doublets within $\text{Cl}(6)$; the result is exact and independent of the chosen embedding, and shows that $\text{Cl}(6)$ is algebraically maximal for QCD. Second, we exhibit the minimal resolution as the graded tensor product $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$: weak isospin acts on the adjoined $\text{Cl}(4)$ factor, the Pati–Salam embedding $\text{Spin}(10) \supset \text{SU}(4) \times \text{SU}(2)_L \times \text{SU}(2)_R$ restores the doublet assignment, and all sixteen charges of one fermion generation are reproduced. This construction is a structural re-expression of known Pati–Salam content; its value is as an explicit bridge among the lattice $\text{Cl}(6)$, the Furey–Gresnigt algebraic, and the Catterall Kähler–Dirac programmes. Third, we prove two no-go statements bounding the dynamics: the diagonal electroweak grade decomposition is inert, equal to one-half at all couplings, and carries no dynamical information; and no parameter of the Higgs potential follows from the algebra, leaving the electroweak scale and Higgs mass undetermined. We clarify, rather than refute, Stoica’s “broken by geometry” characterization: the geometry fixes the breaking *pattern* and embedding ratios, not the breaking *dynamics*.

Keywords: Clifford algebra; $\text{Cl}(6)$; Schur’s lemma; Pati–Salam; $\text{Cl}(10)$; lattice gauge theory; electroweak symmetry breaking; derivability limit; no-go theorem.

1 Introduction

The Clifford algebra $\text{Cl}(6)$ has proven a natural algebraic home for the color sector of the Standard Model. Its eight-dimensional minimal left ideal carries the $\text{SU}(3)_C$

representation content $\mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1} \oplus \mathbf{1}$, and Papers 1 and 2 of this series showed that this structure supports genuine dynamical calculation: Paper 1 [1] verified a Wilson–Dirac operator built in $\text{Cl}(6)$ against sixteen independent benchmarks, and Paper 2 [2] constructed and measured grade-decomposed QCD correlators on quenched lattice ensembles. The natural next question is whether the rest of the Standard-Model gauge structure—in particular weak isospin $\text{SU}(2)_L$ —can be accommodated within the same algebraic setting.

This paper answers that question by establishing where the algebraic machinery reaches and where it stops. We organize the paper around three results. First (Sec. 2), a Schur obstruction: no $\text{SU}(2)$ commuting with color can act on the $\text{Cl}(6)$ ideal except on the color singlets, so quarks cannot be weak doublets within $\text{Cl}(6)$. The result is exact and embedding-independent, and it shows that $\text{Cl}(6)$ is algebraically maximal for QCD. Second (Sec. 3), the minimal resolution: the graded tensor product $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$ adjoins a factor on which weak isospin can act, restoring the doublet assignment through the Pati–Salam embedding and reproducing one full fermion generation. Third (Secs. 4 and 5), two no-go statements that bound the dynamics: the diagonal electroweak grade decomposition is inert at all couplings (Sec. 4), and no parameter of the Higgs potential follows from the algebra (Sec. 5).

These results sit alongside, and sharpen, an existing body of work. Furey and collaborators [3–7] have developed the algebraic Standard-Model programme from division algebras, recovering the gauge group and one generation’s quantum numbers; Stoica [8], in this journal, constructed the electroweak sector in the $\text{Cl}(6)$ language and characterized the electroweak symmetry as “already broken by the geometry of the algebra.” We engage that characterization directly in Sec. 5, where we argue it should be read as a statement about the breaking *pattern* and embedding ratios rather than the breaking *dynamics*—a clarification, not a contradiction.

The paper is therefore a limits note within the series: its contribution is to delimit precisely what the $\text{Cl}(6)$ construction determines, to exhibit the forced extension that lifts its central obstruction, and to bridge the lattice construction of Papers 1–2 to the broader algebraic and lattice programmes it neighbors.

2 The Schur Obstruction

2.1 Setup

Papers 1 and 2 of this series established that the eight-dimensional minimal left ideal of the Euclidean Clifford algebra $\text{Cl}(6)$, with generators satisfying $\{e_i, e_j\} = 2\delta_{ij}$, is the algebraic home of the QCD color sector. The color gauge group $\text{SU}(3)_C$ is embedded through the chain $\text{Spin}(6) \supset \text{SU}(3) \times \text{U}(1)$, and under $\text{SU}(3)_C$ the ideal decomposes into irreducible representations as

$$\mathbf{8} = \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}_a \oplus \mathbf{1}_b. \quad (1)$$

The triplet $\mathbf{3}$ carries the quark color charge; the antitriplet $\bar{\mathbf{3}}$ carries the antiquark charge; and the two color singlets $\mathbf{1}_a, \mathbf{1}_b$ complete the space. This decomposition is fixed by the embedding and is verified at machine precision in Paper 1.

The question this section settles is whether a second non-abelian gauge factor—specifically a weak-isospin $SU(2)_L$ commuting with color, as the Standard Model requires—can be realized on this same eight-dimensional space. The answer is no, and the obstruction is exact and embedding-independent.

2.2 The Commutant of $SU(3)_C$

In the Standard Model, color and weak isospin commute: $[SU(3)_C, SU(2)_L] = 0$. Any candidate $SU(2)_L$ acting on the ideal must therefore have all of its generators in the commutant of the $SU(3)_C$ representation. By Schur’s lemma [9], the commutant of a completely reducible representation is determined entirely by its irreducible content and the multiplicities with which the irreducibles appear.

Reading off the multiplicities from (1):

- The triplet $\mathbf{3}$ appears once.
- The antitriplet $\bar{\mathbf{3}}$ appears once. For $SU(3)$ the fundamental and the antifundamental are *inequivalent* irreducible representations—the fundamental is complex, not self-conjugate—so no nonzero intertwiner maps the $\mathbf{3}$ block to the $\bar{\mathbf{3}}$ block.
- The trivial representation appears twice, as $\mathbf{1}_a$ and $\mathbf{1}_b$. These two copies are equivalent (both trivial), so intertwiners may freely mix them.

Schur’s lemma then gives the commutant as a direct sum of one matrix algebra per equivalence class of irreducible, of size equal to the multiplicity:

$$\text{Comm}(SU(3)_C) = \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}), \quad (2)$$

of total complex dimension $1 + 1 + 4 = 6$. The first two summands are the scalar multiples of the identity on the $\mathbf{3}$ and $\bar{\mathbf{3}}$ blocks respectively; the third is the full algebra of 2×2 complex matrices acting on the two-dimensional space spanned by the color singlets $\{\mathbf{1}_a, \mathbf{1}_b\}$.

2.3 The Obstruction

The Lie algebra of the commutant (2) is

$$\mathfrak{u}(1) \oplus \mathfrak{u}(1) \oplus \mathfrak{u}(2) = \mathfrak{u}(1)^{\oplus 3} \oplus \mathfrak{su}(2). \quad (3)$$

Its only non-abelian simple ideal is the $\mathfrak{su}(2)$ contained in $\mathfrak{u}(2)$. That $\mathfrak{su}(2)$ acts on the two-dimensional space of color singlets and is trivial on every other block. We record the consequence as a theorem.

Theorem 2.1 (Schur Obstruction). *Let $SU(3)_C$ act on the eight-dimensional $\text{Cl}(6)$ ideal by the decomposition (1). Then every $SU(2)$ subgroup commuting with $SU(3)_C$ acts nontrivially only on the two color singlets $\mathbf{1}_a, \mathbf{1}_b$, and acts as the identity on the color triplet $\mathbf{3}$ and antitriplet $\bar{\mathbf{3}}$. Consequently the quark triplet is a singlet under any such $SU(2)$, the exact opposite of the Standard Model assignment in which left-handed*

quarks form weak-isospin doublets. The conclusion depends only on the irreducible content and multiplicities in (1), and is therefore independent of the choice of generator matrices.

Proof. By the commutativity requirement, the generators of any color-commuting $SU(2)$ lie in $\text{Comm}(SU(3)_C)$. By Schur’s lemma this commutant is (2), whose Lie algebra (3) has $\mathfrak{su}(2)$ as its only non-abelian simple ideal. That $\mathfrak{su}(2)$ is supported on the span of $\{\mathbf{1}_a, \mathbf{1}_b\}$ and annihilates the $\mathbf{3}$ and $\bar{\mathbf{3}}$ blocks. Hence any color-commuting $SU(2)$ leaves the triplet invariant. The argument uses only (1), so it holds for every embedding producing that decomposition. $\square$

The inequivalence of $\mathbf{3}$ and $\bar{\mathbf{3}}$ is the load-bearing step: it is what reduces each of those two blocks to a single scalar rather than a larger matrix algebra. Were the representation real or pseudoreal in those blocks, the commutant would be larger and the conclusion could differ; for $SU(3)$ it is not.

2.4 Algebraic Maximality

Corollary 2.2 (Algebraic Maximality for QCD). *The $Cl(6)$ Wilson–Dirac framework of Papers 1 and 2 is algebraically maximal for QCD: $SU(3)_C$ exhausts the nontrivial gauge structure the eight-dimensional ideal can carry while keeping quarks in the color triplet. A nontrivial weak-isospin $SU(2)_L$ acting correctly on quarks cannot be added within $Cl(6)$; it requires enlarging the algebra.*

This reframes what might look like a limitation as a positive structural statement. $Cl(6)$ is exactly the right size for the color sector—neither too small to carry $SU(3)_C$ nor large enough to accommodate a spurious extra non-abelian factor on the quarks. The next section shows that the minimal enlargement consistent with the Standard Model assignment is forced to be $Cl(10)$.

2.5 Numerical Confirmation

The theorem is exact algebra and needs no numerics, but a direct check rules out the most natural attempt to build a color-commuting weak generator by hand. Constructing a candidate T_3 in the eight-dimensional space and testing it against the off-diagonal Gell-Mann generators of $SU(3)_C$ gives

$$\| [T_3, T_a] \| = 1.0 \quad \text{for } a \in \{1, 2, 6, 7\}, \quad (4)$$

i.e. the commutator fails to vanish. This matches the prediction of Theorem 2.1 that no such color-commuting $SU(2)$ generator exists on the triplet. The full check is reproduced in the verification script accompanying this paper.

3 The Forced Extension: $Cl(10) = Cl(6) \hat{\otimes} Cl(4)$

Theorem 2.1 showed that weak isospin has no color-commuting home inside $Cl(6)$: any $SU(2)$ commuting with $SU(3)_C$ is supported on the color singlets and leaves

quarks invariant. The Standard Model requires the opposite—left-handed quarks are weak doublets. The resolution is not a different embedding of $SU(2)_L$ within $Cl(6)$, which Theorem 2.1 rules out for all embeddings, but an enlargement of the algebra itself. This section exhibits the minimal such enlargement that restores the doublet assignment and shows precisely how it sidesteps the obstruction.

3.1 The Graded Tensor Product Construction

We construct $Cl(10)$ as the graded ($\mathbb{Z}_2$ -graded, or “super”) tensor product of $Cl(6)$ with $Cl(4)$,

$$Cl(10) = Cl(6) \hat{\otimes} Cl(4), \quad (5)$$

acting on the 32-dimensional space $\mathbb{C}^8 \otimes \mathbb{C}^4$. Let $\Gamma_7 = i e_1 e_2 \cdots e_6$ denote the $Cl(6)$ chirality element, normalized so that $\Gamma_7^2 = +I_8$, and let $\{f_1, f_2, f_3, f_4\}$ generate $Cl(4)$ with $\{f_j, f_k\} = 2\delta_{jk}$. The ten generators of $Cl(10)$ are

$$\Gamma_i = e_i \otimes I_4 \quad (i = 1, \dots, 6), \quad (6)$$

$$\Gamma_{6+j} = \Gamma_7 \otimes f_j \quad (j = 1, \dots, 4). \quad (7)$$

The factor Γ_7 in (7) is the grading cocycle of the graded tensor product. It is exactly what is needed to make the cross anticommutators vanish: for $i \in \{1, \dots, 6\}$ and $j \in \{1, \dots, 4\}$,

$$\{\Gamma_i, \Gamma_{6+j}\} = (e_i \Gamma_7 + \Gamma_7 e_i) \otimes f_j = 0, \quad (8)$$

because e_i anticommutes with the top-grade element Γ_7 in $Cl(6)$. Without the Γ_7 insertion these would not vanish and (5) would fail. With it, one verifies the full Clifford relation

$$\{\Gamma_a, \Gamma_b\} = 2\delta_{ab} I_{32}, \quad a, b \in \{1, \dots, 10\}, \quad (9)$$

confirmed at machine precision in the verification script accompanying this paper.

The factorization $Cl(6) \otimes Cl(4) \cong Cl(10)$, with the color structure carried by the $Cl(6)$ factor and the weak structure by the $Cl(4)$ factor, is due to Gresnigt [10]; we adopt it directly. Our contribution in this section is the explicit *graded* tensor-product realization with the Γ_7 cocycle (7), which makes the $Cl(10)$ closure manifest, and the use of that realization to lift the obstruction of Theorem 2.1; the underlying $Cl(6) \otimes Cl(4)$ assignment is Gresnigt’s.

3.2 Chirality and the 16-Dimensional Weyl Spinor

The $Cl(10)$ chirality element factorizes cleanly across the two tensor factors:

$$\Gamma_{\text{chiral}}^{Cl(10)} = \Gamma_7^{Cl(6)} \otimes \gamma_5^{Cl(4)}, \quad (10)$$

where $\gamma_5^{Cl(4)} = f_1 f_2 f_3 f_4$ is the $Cl(4)$ chirality. Its $+1$ eigenspace is 16-dimensional and carries exactly one Standard Model generation.

3.3 The Pati–Salam Embedding

The $\text{Spin}(10)$ Lie algebra is spanned by the bivectors $\Sigma_{ab} = \frac{i}{4}[\Gamma_a, \Gamma_b]$. It contains the Pati–Salam subalgebra

$$\text{Spin}(10) \supset \text{SU}(4) \times \text{SU}(2)_L \times \text{SU}(2)_R, \quad (11)$$

under which the 16-dimensional Weyl spinor branches as

$$\mathbf{16} = (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2}). \quad (12)$$

The $(\mathbf{4}, \mathbf{2}, \mathbf{1})$ carries the left-handed fields; the $(\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$ carries the right-handed fields as a charge-conjugate multiplet. This embedding and branching are standard [11, 12]; we use them, and do not claim them as new. What is constructed here is their explicit realization through the graded tensor product (5)–(7).

3.4 How the Enlargement Evades the Schur Obstruction

This is the structural point of the section, and it connects directly to Theorem 2.1. Inside $\text{SU}(4)$ the color group sits as

$$\text{SU}(4) \supset \text{SU}(3)_C \times \text{U}(1)_{B-L}, \quad (13)$$

with $\text{SU}(3)_C$ residing entirely in the $\text{Cl}(6)$ tensor factor: the fundamental $\mathbf{4}$ of $\text{SU}(4)$ is the $\Gamma_7 = +1$ sector of the $\text{Cl}(6)$ construction, and the $B - L$ charge is proportional to the $\text{Cl}(6)$ Cartan generator, $B - L = \frac{2}{3}H$.

Weak isospin $\text{SU}(2)_L$, by contrast, acts on the adjoined $\text{Cl}(4)$ factor—a different tensor slot from color. This is precisely why the obstruction of Theorem 2.1 does not apply. That theorem constrained any $\text{SU}(2)$ lying in the commutant of $\text{SU}(3)_C$ *within* $\text{Cl}(6)$, forcing it onto the color singlets. Here $\text{SU}(2)_L$ does not live in $\text{Cl}(6)$ at all; it lives in the $\text{Cl}(4)$ factor that the tensor product adjoins. Because color acts as I_4 on that factor (6), $\text{SU}(2)_L$ commutes with $\text{SU}(3)_C$ automatically and acts on the full $\text{SU}(4)$ multiplet—and therefore on the quarks—as a genuine doublet symmetry. The tensor enlargement is exactly the minimal step that supplies weak isospin a home where it can act on quarks the way the Standard Model demands.

3.5 Charge Assignments

The electroweak hypercharge and electric charge are recovered from the Pati–Salam data by the standard relations

$$Y = T_{3R} + \frac{1}{2}(B - L), \quad Q = T_{3L} + Y, \quad (14)$$

with T_{3L}, T_{3R} the Cartan generators of $\text{SU}(2)_L, \text{SU}(2)_R$. Evaluating (14) on all sixteen states of one generation—the left-handed doublets $(\nu_L, e_L), (u_L, d_L)$ and the right-handed singlets e_R, u_R, d_R, ν_R —reproduces every Standard Model charge, with residual at most 1.4×10^{-16} and total charge $\sum Q = 0$ over the generation (verification script, 16-charge table).

One subtlety must be stated. The right-handed fields sit in the charge-conjugate block $(\bar{4}, \mathbf{1}, \mathbf{2})$. Consequently $SU(2)_R$ must be represented on that block by $-\tau^*/2$ rather than $+\tau/2$; representing it without the conjugation yields incorrect right-handed charges. This is a representation-theoretic bookkeeping point about the conjugate multiplet, not a physical input, and it is built into the construction.

3.6 Status of the Result

We state plainly what this section is and is not. The Pati–Salam content of $\text{Spin}(10)$ is known and credited above; we make no claim of new physics here, and by the informational-novelty standard applied throughout this series this section is a structural re-expression rather than an informational advance. Its value is constructive and connective: it provides an explicit graded-tensor-product realization $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$ that (i) takes the lattice-ready $\text{Cl}(6)$ framework of Papers 1–2 as a literal tensor factor, (ii) restores the weak-doublet assignment that Theorem 2.1 forbids inside $\text{Cl}(6)$ alone, and (iii) thereby links three otherwise-separate programmes—the lattice $\text{Cl}(6)$ construction of this series, the algebraic Standard-Model work of Furey [6, 7] and Gresnigt [10], and the Kähler–Dirac / Pati–Salam lattice work of Catterall [13, 14]. The extension is “minimal” in the precise sense that it is the smallest graded-tensor enlargement of $\text{Cl}(6)$ restoring the doublet assignment; we do not claim it is the unique conceivable enlargement.

4 Algebraic Rigidity of the T_3 -Diagonal Grade Decomposition

Section 5 will show that the algebra does not constrain the Higgs potential. This section establishes a companion negative result on the gauge side: the diagonal part of the electroweak grade decomposition is dynamically inert. It equals one-half identically—for every gauge configuration and at every coupling—and therefore carries no dynamical information. We state this precisely and prove it, then contrast it with the genuinely dynamical color decomposition of Paper 2, since the two look superficially similar and are not the same kind of object.

4.1 The Decomposition

Working in $SU(2)$ pure gauge theory implemented in the eight-dimensional space of Paper 3’s successor construction, the weak-isospin generators are embedded in the second tensor factor, $T_a = I_2 \otimes (\sigma_a/2) \otimes I_2$, and a gauge link $U \in SU(2)$ is embedded as $I_2 \otimes U \otimes I_2$. The T_3 spectral projectors are

$$P_{\pm}^{\text{EW}} = I_2 \otimes P_{\uparrow/\downarrow} \otimes I_2, \quad (15)$$

with $P_{\uparrow} = (I + \sigma_z)/2$, $P_{\downarrow} = (I - \sigma_z)/2$. Inserting these into the gauge correlator splits it into diagonal and off-diagonal parts,

$$G(t) = G^{(+)}(t) + G^{(-)}(t) + \Delta_{\text{cross}}(t), \quad (16)$$

where $G^{(\pm)}(t)$ are the T_3 -diagonal projected correlators and $\Delta_{\text{cross}}(t)$ is the off-diagonal contribution. The off-diagonal term is the carrier of the non-trivial dynamics; it reduces to $\Delta_{\text{cross}}(t) = 8 \sum_{x,\mu} \text{Re} [b(x, t) \bar{b}(x, 0)]$, where b is the off-diagonal SU(2) link element. We do not develop Δ_{cross} in this paper; it is the subject of later work and is noted again in Sec. 6.

4.2 The Rigidity Result

Define the diagonal grade fractions

$$f_{\pm}(t) = \frac{G^{(\pm)}(t)}{G^{(+)}(t) + G^{(-)}(t)}. \quad (17)$$

Proposition 4.1 (Algebraic Rigidity of the Diagonal Decomposition). *For every gauge configuration, and at all values of the coupling, $f_+(t) = f_-(t) = \frac{1}{2}$. The diagonal grade fractions are constant by construction and contain no dynamical information.*

Proof. The diagonal projectors (15) extract the (1, 1) and (2, 2) matrix elements of the SU(2)-valued line product $W(t) = \prod U$ entering the correlator. Since $W(t) \in \text{SU}(2)$, unitarity and unimodularity give $W_{22} = (W_{11})^*$, hence $|W_{11}| = |W_{22}|$. The diagonal correlators $G^{(\pm)}$ depend on these matrix elements only through their moduli and do so symmetrically, so $G^{(+)}(t) = G^{(-)}(t)$ configuration by configuration, independent of the coupling. The fractions (17) are therefore each $\frac{1}{2}$. The full tensor-index proof, including the rank-one collapse step, is given in Paper 4 of this series. $\square$

The content of Proposition 4.1 is that the equality is exact and unconditional. There are no deviations from $\frac{1}{2}$ to measure: the diagonal decomposition is rigid.

4.3 What This Result Is, and What It Is Not

Proposition 4.1 resembles the color-sector result of Paper 2 only on the surface. The two differ in kind, and conflating them would misrepresent both.

In Paper 2 the color grade fractions satisfy $f_r = \dim(r)/8$ *exactly in the free field* and only *approximately* in the interacting theory; the deviations are genuine and dynamical. The mechanism is traceable: the Cartan grading operator is $H = B_{12} + B_{34} + B_{56}$, and the spacetime Dirac generators $e_1, \dots, e_4$ act on the same tensor factors (qubits 1 and 2) as B_{12} and B_{34} . Dirac / chirality structure therefore feeds the color-grade label, and the interacting-theory deviations record it. That is a kinematic limit theorem with measurable config-by-config departures.

In the present case the diagonal fractions equal $\frac{1}{2}$ at *all* couplings, exactly. This is an algebraic identity, not a kinematic limit theorem, and it must not be called one. The diagonal projection is sensitive only to $|W_{11}| = |W_{22}|$; it is blind to the gauge dynamics by construction. Where the color decomposition has a dynamical signal sitting on top of its free-field value, the diagonal electroweak decomposition has nothing on top of its constant value at all.

We also fix terminology. The object in (16) is the T_3 -diagonal grade decomposition of the $SU(2)$ gauge-link correlator. It is not a “ W -boson correlator”: there is no W boson in pure $SU(2)$ gauge theory absent electroweak symmetry breaking, and no Higgs sector is present here. And, by the above, it is not governed by a kinematic limit theorem. These naming points are not pedantry; the earlier “kinematic theorem” framing obscured exactly the free-versus-all-couplings distinction that distinguishes Proposition 4.1 from the Paper 2 result.

4.4 The No-Go Content

Proposition 4.1 is the electroweak companion to the Derivability Limit of Sec. 5. Both delimit the algebraic machinery from below: the algebra fixes neither the Higgs potential (Sec. 5) nor any dynamics in the diagonal electroweak grade channel (this section). The diagonal grade decomposition is a consistency-bookkeeping device, not a probe of gauge dynamics. The channel that does carry dynamics is the off-diagonal Δ_{cross} of (16), which lies outside the scope of this limits paper and is taken up separately (Sec. 6).

5 The Derivability Limit

5.1 Statement and Proof

We state the central negative result on the scalar sector.

Theorem 5.1 (Derivability Limit). *No condition on the Higgs potential*

$$V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4 \tag{18}$$

follows from the algebraic structure of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ alone.

Proof. The algebraic structure encodes three things: the representation content of the fundamental fields, the action of the gauge group on those representations, and the charge assignments and symmetry relations among them. It does not encode a choice of Lagrangian.

The Higgs field transforms as the doublet $(\mathbf{2}, \frac{1}{2})$ of $SU(2)_L \times U(1)_Y$. For a single such doublet, $\phi^\dagger \phi$ is the unique quadratic gauge invariant, and every positive power of it is likewise invariant. Hence for any $\mu^2 \in \mathbb{R}$ and any $\lambda > 0$, the potential (18) is fully invariant under $SU(2)_L \times U(1)_Y$: the algebraic structure is consistent with this choice of dynamics.

Since the algebra is consistent with the entire two-parameter family $\{(\mu^2, \lambda) : \mu^2 \in \mathbb{R}, \lambda > 0\}$, it constrains no member of it. In particular it is consistent with $\mu^2 > 0$ (unbroken symmetry), $\mu^2 = 0$ (massless tree-level scalar), and $\mu^2 < 0$ (the physical broken case), and with any magnitude of λ . The algebra constrains neither the sign of μ^2 nor the value of λ . $\square$

Corollary 5.2. *The electroweak scale $v = \sqrt{-\mu^2/\lambda}$ is unconstrained by the algebra; in particular the algebra does not explain why v is of order 246 GeV rather than the Planck scale.*

Corollary 5.3. *The Higgs boson mass $m_H^2 = 2\lambda v^2 = -2\mu^2$ is unconstrained by the algebra.*

5.2 What the Theorem Is: A Precise Statement of Standard Lore

We are deliberate about the status of Theorem 5.1, because it is easy to overstate. At its core it is the standard effective-field-theory fact that a symmetry determines the *form* of the operators it permits, not their *coefficients* [15]. Gauge invariance tells us that $|\phi|^2$ and $|\phi|^4$ are admissible terms; it is silent on μ^2 and λ . This is not a new principle, and we do not present it as one.

What is worth stating precisely is its consequence *within* the division-algebra programme. The programme’s appeal is that representation content and charge structure emerge from the algebra with little external input, and it is natural to ask how far that reach extends. Theorem 5.1 marks the boundary exactly: the reach extends to the symmetry, the representations, and the charges, and stops at the dynamics that fix the scale of breaking. The value of the theorem is as a boundary marker, made precise in this setting and engaged against a specific prior claim in the next subsection. It is not a central discovery of the paper, and a reader who takes it for one has misread our intent.

5.3 Relation to Stoica: Clarification, Not Refutation

Stoica [8], in the same journal this series targets, constructs the electroweak sector in the $\text{Cl}(6)$ language and states that the electroweak symmetry is “already broken by the geometry of the algebra,” reading off a bare Weinberg angle $\sin^2 \theta_W = 0.25$. Theorem 5.1 says the algebra does not fix the Higgs potential. These two statements are sometimes read as being in tension. They are not, and the distinction is instructive.

Stoica’s $\sin^2 \theta_W = 0.25$ is a tree-level generator-normalization ratio. It reflects how the $\text{U}(1)_Y$ generator is normalized relative to the $\text{SU}(2)_L$ generators within the algebraic embedding—a ratio of Cartan normalizations, fixed by the representation content. It is *not* a prediction of the potential parameters μ^2 and λ , and it is not derived from any dynamics of symmetry breaking. Theorem 5.1 therefore does not refute it: the two statements concern different objects. The Weinberg-angle ratio is exactly the kind of quantity the algebra *does* fix (a normalization built into the embedding); the potential is exactly the kind it does *not*.

What Theorem 5.1 does is clarify the phrase “broken by geometry.” The geometry of the algebra fixes the symmetry group, the representation content, and the embedding ratios that yield a tree-level Weinberg angle. It does not fix the electroweak *breaking* itself: the sign of μ^2 that selects the broken vacuum, and the scale v at which breaking occurs, are not determined by the geometry. “Broken by geometry” is apt for the

pattern of symmetry breaking—which fields can acquire vacuum expectation values, and into which unbroken subgroup the group descends—but it overstates the reach if read as fixing the breaking dynamics. The precise division is: geometry fixes the pattern and the representation-theoretic ratios; the potential that triggers and scales the breaking is separate input. We offer this as a refinement of Stoica’s framing, not a contradiction of his result.

5.4 What the Algebra Does Fix: Operator Forms

The Derivability Limit is precise, not total. It is worth recording the complementary positive statement, because it shows the boundary is sharp rather than a blanket disclaimer.

After symmetry breaking, fermion masses arise from Yukawa couplings of the form

$$\mathcal{L}_{\text{Yukawa}} = -y_f \bar{\psi}_L \phi \psi_R + \text{h.c.} \quad (19)$$

The *form* of this coupling is algebraically forced: $\bar{\psi}_L \phi \psi_R$ is the unique renormalizable gauge-invariant bilinear contracting a left-handed doublet, the Higgs doublet, and a right-handed singlet, with all charges summing to zero. Gauge invariance, which the algebra supplies, permits no other renormalizable Yukawa structure. What the algebra does not fix is the *magnitude* y_f . The Standard Model carries thirteen independent Yukawa couplings, spanning six orders of magnitude from the electron to the top quark, and all thirteen are free parameters from the algebraic point of view.

This is the same form-versus-coefficient distinction as Theorem 5.1, now on the Yukawa sector: the algebra determines which couplings may appear and forbids the rest, but assigns none of their values. The limit is a clean line, not an absence of content.

5.5 Remark: The One Speculative Handle

Remark 5.4. There is exactly one route by which a dynamical mechanism, rather than pure representation theory, might constrain μ^2 within the programme. If the Higgs were a composite object—a fermion condensate, as in technicolor, rather than an elementary scalar—then a Banks–Casher-type relation [16] connecting a non-zero condensate to the spectral density of a fundamental operator at zero eigenvalue could in principle constrain the breaking, and the programme’s spectral-projector methods would be well suited to posing the question. This is the right question if the Higgs is composite and the wrong one if it is elementary; the discovery of a Higgs boson with properties consistent with an elementary scalar disfavors it. We flag this as the sole caveat that could alter the verdict of Theorem 5.1 and do not pursue it.

6 Discussion and Outlook

6.1 What the Algebraic Machinery Determines

Taken together, the results of this paper draw a sharp line around the algebraic programme. On one side lie the quantities the algebra fixes. It fixes the representation content: which irreducible representations of the Standard Model gauge group appear, recovered for the color sector from the $\text{Cl}(6)$ ideal of Papers 1 and 2 and for one full generation from the $\text{Cl}(10)$ construction of Sec. 3. It fixes the charge assignments, with the Gell-Mann–Nishijima relation $Q = T_3 + Y$ satisfied by all sixteen states of a generation. It fixes the pattern of symmetry breaking once a vacuum direction is chosen, and the embedding ratios that yield a tree-level Weinberg angle, in the precise sense clarified in Sec. 5.3. It fixes the form of the allowed interactions: the unique gauge-invariant Yukawa bilinear and quartic scalar terms, as set out in Sec. 5.4. And it places the construction at a definite algebraic level: $\text{Cl}(6)$ is maximal for QCD (Corollary 2.2), and $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$ is the minimal graded-tensor extension that restores the weak-doublet assignment Theorem 2.1 forbids within $\text{Cl}(6)$ alone.

A separate, more modest positive should be recorded for what it is. The per-configuration completeness identity $\sum_r C^{(r)} = C$, satisfied to order 10^{-16} on every gauge configuration, is a genuine consistency check on the propagator and projector code. It is a diagnostic, not a physical result, and we claim nothing more for it.

6.2 What It Does Not Determine

On the other side lie the no-go results. The algebra does not fix the Higgs potential—neither the sign of μ^2 nor the magnitude of λ (Theorem 5.1)—and therefore fixes neither the electroweak scale v (Corollary 5.2) nor the Higgs mass m_H (Corollary 5.3). It does not fix the thirteen Yukawa magnitudes or the fermion mass hierarchy they encode (Sec. 5.4). The diagonal electroweak grade channel carries no dynamics at all: its grade fractions are pinned to one-half by construction, at every coupling (Proposition 4.1). The algebra does not select left over right: both $\text{SU}(2)_L$ and $\text{SU}(2)_R$ are present in the Pati–Salam structure of Sec. 3.3, and gauging only the left copy—the parity violation of the physical world—is an external input the algebra accommodates but does not supply. And, as Theorem 2.1 establishes, no color-commuting weak isospin lives inside $\text{Cl}(6)$ in the first place.

6.3 The Bridge Between Three Programmes

Beyond delimiting the machinery, the construction of Sec. 3 serves a connective purpose. The explicit graded-tensor realization $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$ takes the lattice-ready $\text{Cl}(6)$ framework of this series as a literal tensor factor, and in doing so links three otherwise separate lines of work: the lattice $\text{Cl}(6)$ construction developed here, the algebraic Standard-Model programme of Furey [6, 7] and Gresnigt [10], and Catterall’s Kähler–Dirac route to Pati–Salam on the lattice [13, 14]. This bridge is

the paper’s positive role alongside its two no-go results, and it locates the lattice programme within the broader algebraic landscape rather than apart from it.

6.4 Outlook

The line drawn here is not the end of the programme but its honest boundary. The diagonal grade channel is inert, but the off-diagonal cross-channel Δ_{cross} introduced in Sec. 4 is coupling-sensitive and is the natural carrier of gauge dynamics in this decomposition; its analysis, including the gauge-fixing systematics it entails, is taken up elsewhere. That channel, and the composite-Higgs question flagged in Remark 5.4, are the two directions in which the boundaries established in this paper might eventually be pushed.

7 Conclusions

This paper has drawn a precise boundary around what the $\text{Cl}(6)$ -based algebraic programme determines, through three results.

First, a Schur obstruction (Theorem 2.1): no weak isospin commuting with color can act on the eight-dimensional $\text{Cl}(6)$ ideal except on the color singlets, so quarks cannot be weak doublets within $\text{Cl}(6)$. The obstruction is exact and embedding-independent, and it establishes that $\text{Cl}(6)$ is algebraically maximal for QCD (Corollary 2.2).

Second, the minimal resolution (Sec. 3): the graded tensor product $\text{Cl}(10) = \text{Cl}(6) \hat{\otimes} \text{Cl}(4)$ adjoins a $\text{Cl}(4)$ factor on which weak isospin acts, restoring the doublet assignment via the Pati–Salam embedding and reproducing all sixteen charges of one generation. This construction is a structural re-expression of known Pati–Salam content, and its value is as an explicit bridge among the lattice $\text{Cl}(6)$, Furey–Gresnigt, and Catterall programmes.

Third, two no-go statements bounding the dynamics: the diagonal electroweak grade decomposition is inert, equal to one-half at all couplings (Proposition 4.1); and the algebra fixes no parameter of the Higgs potential, hence neither the electroweak scale nor the Higgs mass (Theorem 5.1). Together they mark, precisely, where algebraic structure ends and dynamics begins.

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